Precasterz

Innovative Precast Solutions for Modern Engineering

Design Reports · Shop Drawings · Structural Analysis · Automation

GROUP SPECIALIST ENGINEERS

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ABOUT PRECASTERZ

Precasterz

Precasterz is a team of dedicated Group Specialist Engineers delivering innovative, efficient, and sustainable precast concrete solutions for the construction industry. Founded in 2024, we combine deep structural engineering expertise with advanced software tools to produce precise, fabrication-ready deliverables at every stage of a project.

We specialize in adapting any structural design to an optimized precast solution — covering detailed design reports, general arrangement drawings, shop drawings, BOQ, BBS, proposal preparation, and bespoke engineering automation tools. Our team continuously embraces the latest engineering software to remain at the forefront of the precast industry.

PRECISION	INNOVATION	EFFICIENCY	COLLABORATION
Every drawing and calculation delivered to the highest accuracy standard.	Advanced engineering tools and automation for smarter project outcomes.	Optimized workflows that accelerate delivery without compromising quality.	Working closely with engineers, contractors, and fabricators worldwide.

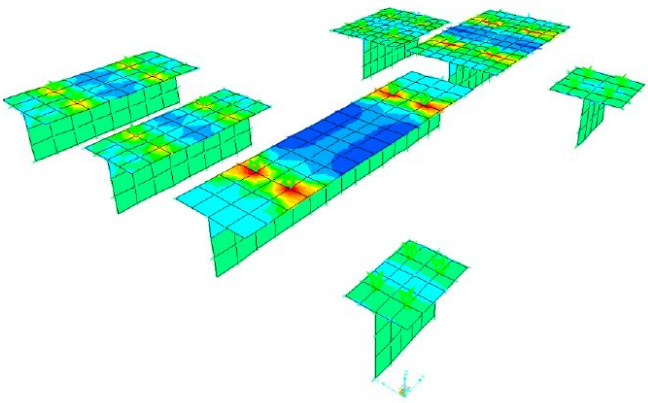
OUR SERVICES OVERVIEW

Precasterz offers a comprehensive range of precast engineering services — from initial proposal through to fabrication-ready shop drawings and quantity reporting. All deliverables are tailored to the specific technical requirements of each project.

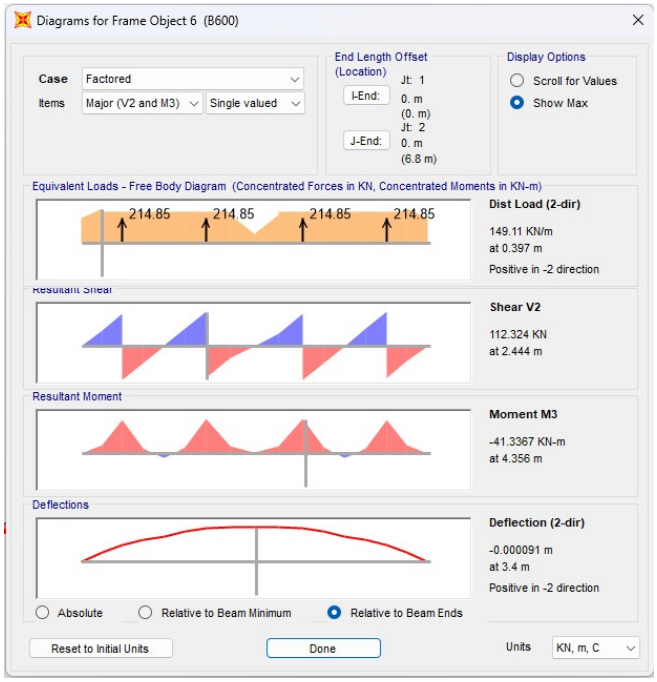
Design Reports & Structural Analysis	Design reports, structural calculations, code compliance (ACI / Eurocode)
Detailed Design / GA Drawings	General arrangement drawings for all precast element types and assemblies
Shop Drawings	Manholes, foundations, hollow-core, boundary walls, building elements
Proposal Preparation	Technical proposals with cost estimates, scope, and specifications
BOQ & Bar Bending Schedules	Precise quantity take-offs and reinforcement schedules for all elements
Engineering Automation	ETABS load combination scripts, Concise Beam autoprinter, custom tools

DESIGN REPORTS, STRUCTURAL ANALYSIS & SAFETY COMPLIANCE

Precasterz delivers comprehensive design reports for all precast element types. Each report includes material specifications, applied loads, sectional analysis, and compliance checks per ACI 318, EN 1992, or project-specific codes. Structural analysis covers element capacities, connection adequacy, deflection, and crack width verification.



FEA stress distribution — precast sleepers



SAP2000 beam force diagrams — factored load case

Structural Design Reports

Full documentation covering material properties, applied loads, sectional analysis, and compliance summaries for all precast element types — columns, beams, slabs, walls, manholes, and foundations.

Structural Analysis & Verification

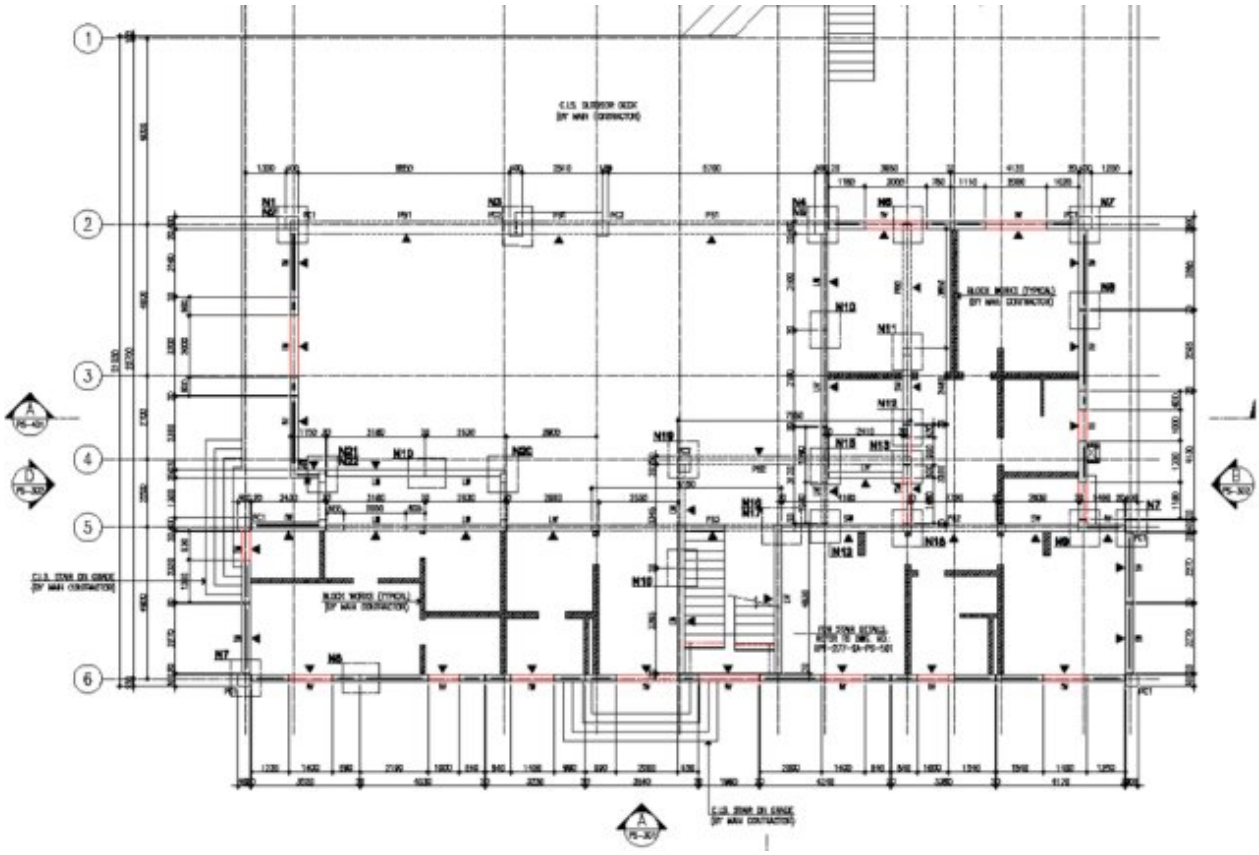
Independent structural checks verifying element capacities, connection adequacy, and global stability. Includes interaction diagrams, deflection checks, and crack width verification.

Safety & Code Compliance

Systematic review against project-specific safety standards and construction regulations. Documentation suitable for client submission, third-party review, and authority approval.

DETAILED DESIGN / GENERAL ARRANGEMENT DRAWINGS

We produce precise general arrangement drawings defining the geometry, positioning, and assembly of precast elements within the overall structure. These are the primary reference for manufacturing, logistics, and site installation — prepared in AutoCAD to client standards.



Precast building floor plan — GA layout with element marks and grid references

General Arrangement Drawings

Element positions, grid references, levels, joint details, and interface conditions. Full title block, revision control, and AutoCAD-native output.

Element Layout Plans & Elevations

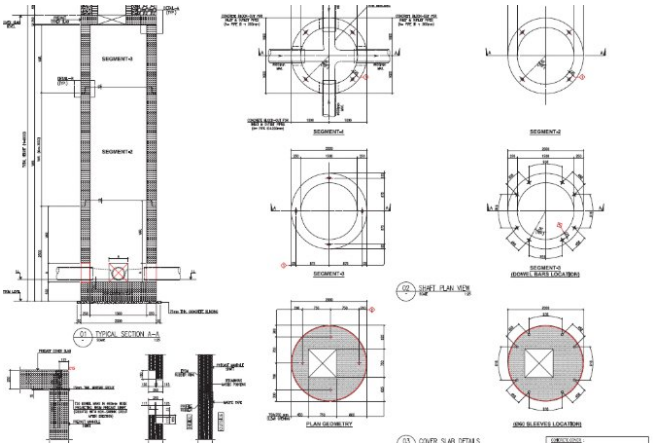
Layout plans and elevations identifying each precast unit by mark number, orientation, and clearance — coordinated against structural and architectural drawings.

Connection & Interface Details

Connection drawings covering bearing conditions, grouting pockets, anchor plates, inserts, and tolerances — ensuring buildability and structural continuity.

SHOP DRAWINGS

Our shop drawings provide complete fabrication information for each precast element — geometry, reinforcement layout, embedded items, lifting inserts, and production notes. We cover all standard precast element types used in civil infrastructure and building construction.



Precast manhole shop drawing — segments and plan views

Fabricated manhole segment — ready for delivery

Manholes & Chambers

Shop drawings including invert levels, wall thicknesses, reinforcement arrangement, step iron positions, lifting points, and sealing details. Designed to suit crane capacity and transport limits.

Precast Foundations

Foundation shop drawings covering geometry, rebar layout, starter bar positions, lifting inserts, and embedded items. Fully coordinated with structural design.

Hollow-Core Slabs

Shop drawings detailing strand arrangement, core configurations, opening locations, core fill zones, and bearing details based on the received structural design.

Boundary Wall Elements

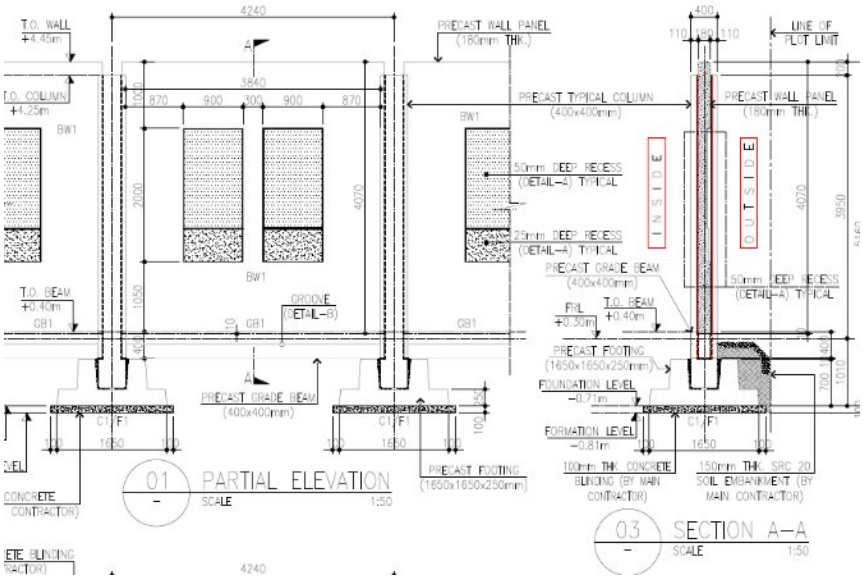
Complete shop drawings for all boundary wall components: columns, beams, wall panels, and footings, including reinforcement, connection details, and surface finishes.

Building Precast Elements

Shop drawings for structural building components — columns, beams, staircases, and façade panels with full reinforcement, connections, and tolerances per design specifications.

PROPOSAL PREPARATION

Precasterz assists clients and precast manufacturers in preparing detailed, technically rigorous project proposals. Our proposals clearly define scope, methodology, technical requirements, and cost basis — providing a solid foundation for tendering and client decision-making.



Precast boundary wall — partial elevation & section A-A

Technical Scope Definition

Structured breakdown of precast scope covering element types, quantities, design standards, interface requirements, and project-specific technical constraints.

Precast Design Methodology

Description of the proposed design approach, applicable codes, analysis methods, and material specifications — demonstrating technical competency to clients and reviewers.

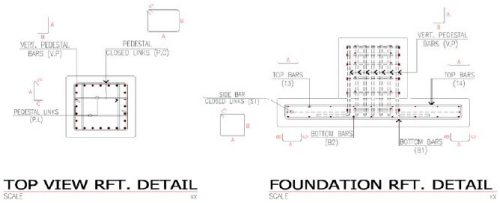
Cost Estimation Support

Assistance with material take-offs and preliminary quantities to support cost estimation, subcontractor pricing, and project budget planning.

ENGINEERING AUTOMATION & QUANTITY SERVICES

Precision meets efficiency. We provide meticulous quantity reporting alongside bespoke automation tools designed to streamline structural design and documentation workflows without compromising technical accuracy.

In addition, we develop custom AutoCAD commands and scripts to automate the production of shop drawings — significantly reducing drafting time, eliminating repetitive manual tasks, and ensuring consistency across all drawing sets.

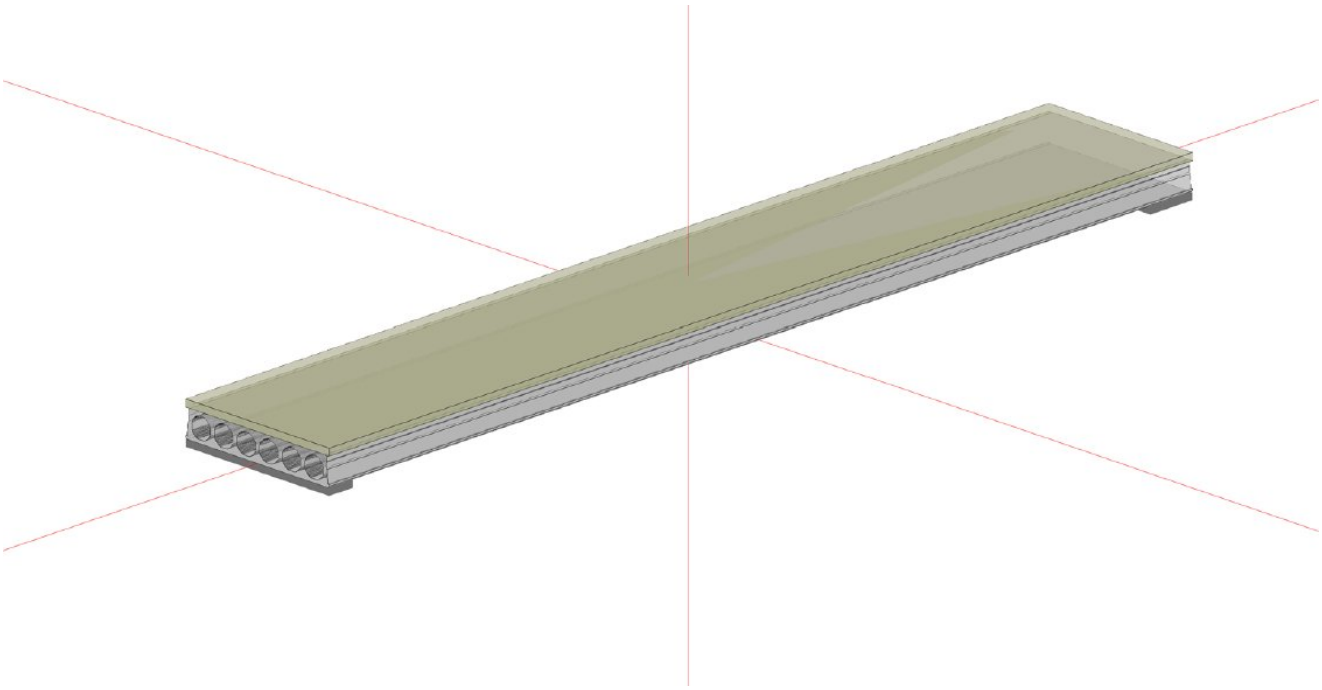


Bar ID	Bar Diameter (mm)	Quantity (Nos.)	A (mm)	B (mm)	C (mm)	Total Length (mm)	Bar Weight (kg/m)	Total Bar Weight (kg)
B1	16	11	2050	250	250	2550	1.580	44.3
B2	16	11	1850	235	235	2320	1.580	40.3
T3	12	11	1850	0	0	1850	0.889	18.1
T4	12	11	2050	0	0	2050	0.889	20.0
S1	12	1	2034	1834	110	7956	0.889	7.1
V.P.1	16	24	1720	300	0	2020	1.580	76.6
P.1.C.1	12	10	500	500	110	2220	0.889	19.7
P.1.C.2	12	10	500	350	110	1920	0.889	17.1
P.1.C.3	12	10	350	500	110	1920	0.889	17.1
							Total Steel (kg)	260.3

Bar bending schedule — precast foundation reinforcement detail

NAME	TYPE	IS AUTO	LOAD NAME	MODE	SF	GUID	NOTES
UL-Combination-1	Linear Add	No	LD		1.4		
UL-Combination-1			SCREED		1.4		
UL-Combination-1			BATTERY		1.4		
UL-Combination-1			ELECTRICAL		1.4		
UL-Combination-2	Linear Add	No	LD		1.2		
UL-Combination-2			SCREED		1.2		
UL-Combination-2			BATTERY		1.2		
UL-Combination-2			ELECTRICAL		1.2		
UL-Combination-2			Lfirst		1.6		
UL-Combination-2			Lsecond		1.6		
UL-Combination-2			Lroof		0.5		
UL-Combination-3	Linear Add	No	LD		1.2		
UL-Combination-3			SCREED		1.2		
UL-Combination-3			BATTERY		1.2		
UL-Combination-3			ELECTRICAL		1.2		
UL-Combination-3			Lroof		1.6		
UL-Combination-3			Lfirst		1		
UL-Combination-3			Lsecond		1		
UL-Combination-4	Linear Add	No	LD		1.2		
UL-Combination-4			SCREED		1.2		
UL-Combination-4			BATTERY		1.2		
UL-Combination-4			ELECTRICAL		1.2		
UL-Combination-4			Lroof		1.6		
UL-Combination-4			W		0.8		
UL-Combination-5	Linear Add	No	LD		1.2		
UL-Combination-5			SCREED		1.2		
UL-Combination-5			BATTERY		1.2		
UL-Combination-5			ELECTRICAL		1.2		
UL-Combination-5			Lroof		1.6		

ETABS load combination automation output



Prestressed hollow-core slab — 3D fabrication model

Bills of Quantities (BOQ) & Bar Bending Schedules (BBS)

Detailed quantity take-offs and bar bending schedules for all precast elements. Output formatted for procurement, cost control, and reinforcement fabrication. Covers manholes, foundations, slabs, walls, and building structures.

ETABS Load Combination Automation

Custom-developed scripts that automate the generation of complex load combinations within ETABS — eliminating manual entry errors and significantly reducing analysis preparation time.

Concise Beam Autoprinter

An innovative batch-processing tool for automated printing of Concise Beam design reports. Eliminates repetitive manual export tasks across multiple beam elements.

Lifting Analysis & Verification

Structural verification of lifting arrangements including sling angles, insert capacity, dynamic amplification factors, and element stress checks during erection — fully documented in compliance with project safety requirements.

CONTACT US

We welcome enquiries from precast manufacturers, main contractors, structural engineers, and project owners. Reach out to discuss your project requirements — we respond promptly.

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